High-Performance In-Memory Data Wrangling using data.table

Week 2 Project Report

Student Details

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Course: Bachelor of Computer Applications (BCA)
Language: R
Package Used: data.table

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1. Project Objective

The objective of this project is to implement high-performance data processing on large-scale transactional datasets using the **data.table** package in R.

The project demonstrates efficient data generation, cleaning, filtering, indexing, aggregation, business analysis, and visualization using a synthetic sales dataset containing **500,000 transaction records**.

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2. Software Used

- R Programming
- Google Colab / RStudio
- data.table Package
- Git and GitHub

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3. Dataset Description

A synthetic enterprise sales transaction dataset was generated using R.

Dataset Name:

sales_transactions.csv

Dataset Size:

- 500,000 transaction records

Features Include:

- TransactionID
- CustomerID

- CustomerName
- ProductID
- ProductName
- Category
- Quantity
- UnitPrice
- Discount
- GST
- PaymentMethod
- City
- State
- OrderDate
- SalesPerson

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4. Implementation Process

Dataset Generation

- Generated 500,000 synthetic sales records.
- Stored the dataset as a CSV file.

Data Exploration and Cleaning

Performed:

- Dataset loading using fread()
- Structure and summary analysis
- Missing value checking
- Duplicate removal
- Data type conversion

Data Manipulation

Implemented:

- Column selection

- Row filtering
- Sorting operations
- Finding top expensive products
- Finding highest quantity transactions

In-Memory Column Updates

Created new columns using := operator:

- Revenue
- DiscountAmount
- GSTAmount
- NetRevenue

Indexing

Used setkey() for faster searching based on:

- CustomerID
- ProductID

Aggregation

Generated reports:

- Customer-wise summary
- Product-wise summary
- City-wise summary
- Category-wise summary

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5. Business Analysis

The project analyzed:

- Highest revenue generating city
- Best performing product
- Highest ordering customer
- Most used payment method
- Average transaction value

- Highest revenue category
- Top 20 customers by revenue
- Top 20 products by quantity sold

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6. Visualization

Generated Base R visualizations:

1. Transactions by City

city_transactions.png

2. Revenue by Category

category_revenue.png

3. Top 20 Customers

top_customers.png

4. Payment Method Distribution

payment_distribution.png

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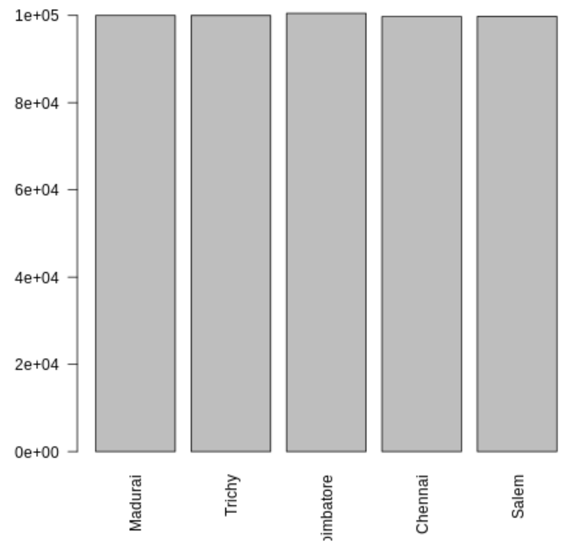
7. Output Files

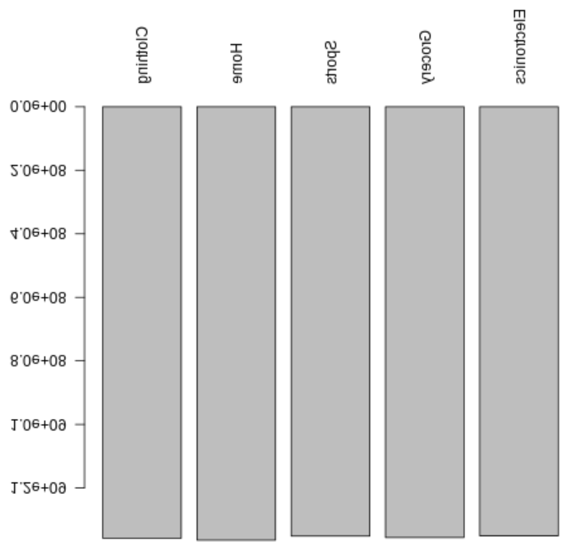
The Output folder contains:

Output/

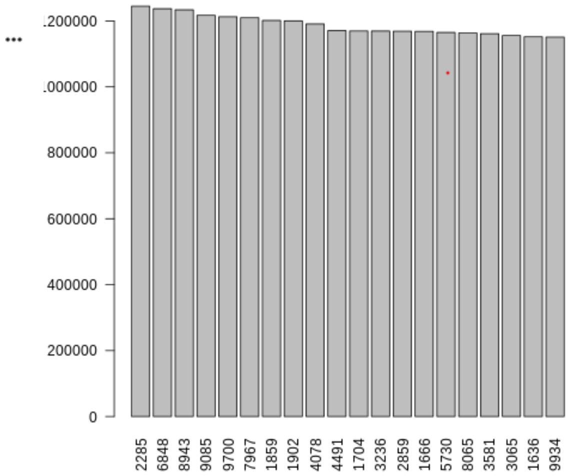
├── city_transactions.png



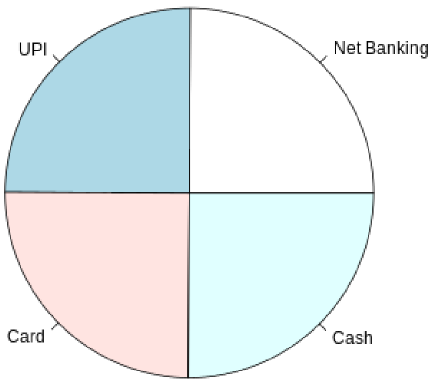
├── category_revenue.png



├── top_customers.png



├── payment_distribution.png



├── customer_summary.csv

├── product_summary.csv

├── city_summary.csv

└── category_summary.csv

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8. Learning Outcomes

Through this project, I learned:

- Large-scale data processing using data.table
- Memory-efficient data manipulation
- Reference semantics using :=
- Data indexing with setkey()

- Data aggregation techniques
- Business analytics and visualization

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9. Conclusion

This project demonstrates how the **data.table** package can efficiently handle large transactional datasets with improved speed and memory management.

The implemented pipeline successfully processes 500,000 records and generates useful business insights through reports and visualizations.